

Kernel 1

q_{16}, k_{16}
(d = 16)



parallel (all CTAs)

D16 block scoring
per-block (m^b, ℓ^b)



histogram

256 bins, 1 KB



sequential (last CTA)

threshold scan

→ block_mask

Grid(B_{size}, N_b)

atomicAdd per CTA

last CTA per head

block_mask

1 0 0 1 0 0 0 1 0 1

Kernel 2

q, k, v
(d = 128)



Select



parallel (all splits)

**block-sparse D128
attention**
online softmax

sequential (last CTA)

**cross-split
reduce**



output

(B, H, 1, 128)

Grid(BH, split_kv), 1 warp / CTA



Sort-free top-p



Bandwidth-optimized